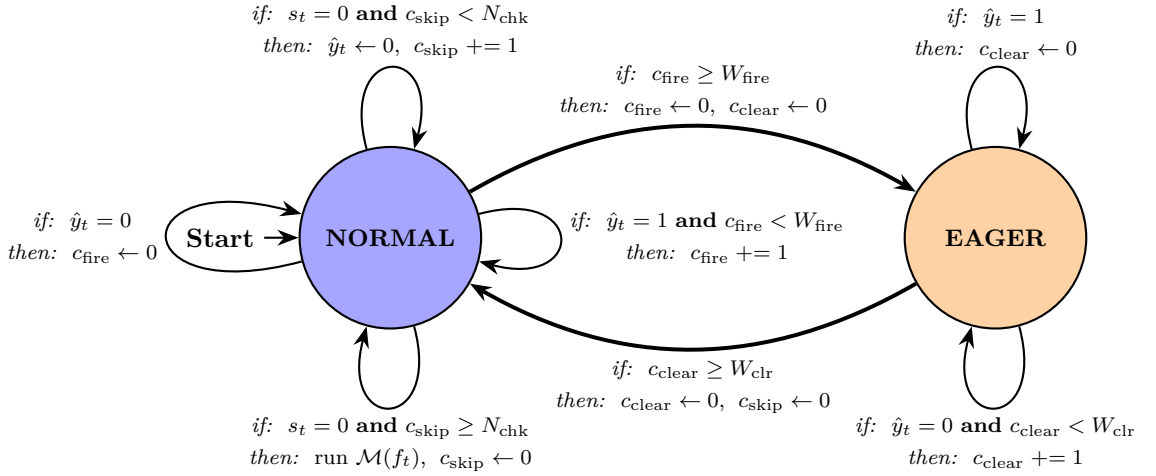




Symbol / Mode	Description
● NORMAL	Skip module $\mathcal{S}$ is active : static frames are skipped; $\mathcal{M}$ is invoked only when motion is detected or $c_{\text{skip}} \geq N_{\text{chk}}$
● EAGER	Skip module $\mathcal{S}$ is bypassed : $\mathcal{M}$ runs on <i>every</i> frame to prevent missed detections
$s_t \in \{0, 1\}$	Skip decision from $\mathcal{S}$: $s_t=0$ no motion detected (skip candidate); $s_t=1$ motion detected (run inference)
$\hat{y}_t \in \{0, 1\}$	Prediction from $\mathcal{M}$: $\hat{y}_t=1$ fire/smoke; $\hat{y}_t=0$ negative
c_{skip}	Consecutive skipped frames (reset to 0 on any inference call)
c_{fire}	Consecutive fire/smoke predictions in NORMAL mode (reset to 0 on any negative)
c_{clear}	Consecutive negative predictions in EAGER mode
W_{fire}	Fire streak length required to enter EAGER mode
W_{clr}	Clear streak length required to exit EAGER mode
N_{chk}	Maximum consecutive skips before a forced inference check